Mobile Equipment (ME)

The Mobile Equipment is the physical mobile device (hardware) used by the subscriber. It includes the radio transmitter/receiver, processor, antenna, microphone, speaker, and display. The ME communicates with the BTS using radio signals over the air interface (Um interface). It does not permanently store subscriber identity; instead, it reads it from the SIM. However, it stores the IMEI (International Mobile Equipment Identity), which uniquely identifies the device itself. When a user makes a call, sends SMS, or uses data, the ME converts voice into digital signals and sends them to the BTS. It receives encrypted signals from the network and converts them back into voice or data. If a phone is reported stolen, the IMEI can be checked in the EIR to block it. The ME's main purpose is to provide the physical interface between the human user and the GSM network.

Subscriber Identity Module (SIM)

The SIM card is a smart card that stores subscriber-specific information. It holds the IMSI (International Mobile Subscriber Identity), authentication key (Ki), SMS storage, contacts (optional), and service-related data. The SIM does not directly transmit signals; it works together with the ME. During network registration, the SIM sends the IMSI to the network for identification. The Authentication Center (AuC) verifies the subscriber using the secret key stored in both the SIM and AuC. The SIM enables mobility — you can remove it and insert it into another phone and still access your services. It receives authentication challenges from the network and calculates responses. Its main purpose is secure subscriber identification and service authorization.

Base Transceiver Station (BTS)

The BTS is the radio communication tower that directly communicates with mobile phones. It transmits and receives radio signals to and from the Mobile Station. The BTS does not store permanent subscriber data but temporarily processes call and signaling information. When a user initiates a call, the BTS receives the radio signal and forwards it to the BSC. It also sends paging messages to locate subscribers for incoming calls. BTS handles channel coding, encryption at the radio level, and signal modulation. In handover situations (when a user moves), the BTS measures signal strength and reports to the BSC. Its purpose is to provide wireless connectivity between users and the GSM network.

Base Station Controller (BSC)

The BSC controls multiple BTS units and manages radio resources. It allocates frequency channels, controls handovers between BTS cells, and manages traffic load. It receives voice/data signals from BTS and forwards them to the MSC. It temporarily holds call-related data such as

channel assignments and signal quality reports. During mobility (when a user moves between cells), the BSC decides when to switch the connection to another BTS. It ensures efficient use of limited radio spectrum. The BSC acts like a traffic manager coordinating multiple BTS towers. Its main purpose is radio network management and mobility control.

Mobile Switching Center (MSC)

The MSC is the central switching system of the GSM core network. It connects calls between mobile users and between GSM and external networks. It does not permanently store subscriber profiles but retrieves them from HLR/VLR when needed. When a call is made, the MSC checks subscriber permissions and routes the call to the correct destination. It coordinates with the VLR to confirm the user's current location. It also manages SMS routing and call setup/termination. In roaming situations, the MSC communicates with other MSCs. The MSC's purpose is call control, switching, and mobility management.

Visitor Location Register (VLR)

The VLR is a temporary database linked to an MSC. It stores information about subscribers currently within that MSC's coverage area. It receives subscriber data from the HLR when a user enters a new location area. The VLR stores IMSI, MSISDN, service permissions, and location area identity. It reduces the need to frequently contact the HLR. When a call arrives, the MSC queries the VLR to confirm the subscriber's status. Once the subscriber leaves the area, the data may be deleted. Its purpose is temporary subscriber management during roaming.

Home Location Register (HLR)

The HLR is the main database that permanently stores subscriber information. It holds IMSI, MSISDN (phone number), subscribed services, billing details, and current VLR location. When a user registers in a new area, the HLR updates the location by storing the new VLR address. For incoming calls, the MSC queries the HLR to determine where the subscriber is located. The HLR sends necessary subscriber data to the serving VLR. It acts like a national citizen registry for mobile subscribers. Its purpose is permanent subscriber data storage and global location tracking.

Authentication Center (AuC)

The AuC provides security functions in GSM. It stores the secret authentication key (Ki) for each subscriber, copied from the SIM during registration. When a subscriber tries to connect, the AuC generates a random challenge (RAND). The SIM calculates a response (SRES), and the AuC

verifies it. If correct, access is granted. The AuC also generates encryption keys (Kc) for secure communication. It does not handle calls directly but supports authentication processes. Its purpose is to prevent unauthorized network access and ensure secure communication.

Equipment Identity Register (EIR)

The EIR stores information about mobile device IMEI numbers. It contains three lists: white (allowed devices), grey (monitored devices), and black (stolen/blocked devices). When a device connects, the MSC can check the IMEI with the EIR. If the IMEI is blacklisted, service is denied even if the SIM is valid. The EIR helps prevent stolen phone usage. It does not store subscriber data — only equipment data. Its purpose is device control and network security.

Gateway Mobile Switching Center (GMSC)

The GMSC acts as the entry and exit point between the GSM network and external networks like PSTN or other mobile networks. When someone from outside GSM calls a GSM subscriber, the call first reaches the GMSC. The GMSC queries the HLR to find the subscriber's current MSC location. It then routes the call to the correct MSC. It does not permanently store subscriber data. Its purpose is inter-network call routing and connectivity.

Public Switched Telephone Network (PSTN)

The PSTN is the traditional landline telephone network. It carries voice calls using circuit switching. When a GSM user calls a landline, the MSC routes the call through the GMSC to the PSTN. It does not store GSM subscriber data. It only carries voice traffic. Its purpose is to allow communication between mobile users and fixed-line users worldwide.

Public Land Mobile Network (PLMN)

A PLMN is any mobile network operated by a telecom provider. GSM networks are examples of PLMN. When roaming internationally, your home PLMN communicates with a visited PLMN. They exchange subscriber data through signaling systems. The visited PLMN temporarily stores subscriber data in its VLR. Its purpose is to enable inter-operator communication and roaming services.

Internet

The Internet connects GSM users to global data services. Through technologies like GPRS or EDGE, GSM networks connect to IP-based networks. Data requests from a mobile device pass through the MSC and packet core network to Internet servers. It does not store subscriber identity but may use IP addressing. Its purpose is to provide data services like browsing, email, and online communication.